

MINDBIT SOLUTIONS LLP · ROOT PLATFORM · CONFIDENTIAL

Software Requirements Specification

Guest House Operations Hub — ROOT Platform PMS Layer

Functional and non-functional requirements with build status, binding constraints, external interfaces, data model, error handling, security, performance and observability. Structured per IEEE 29148/830 principles.

Document	SRS-GHOH-001 · Software Requirements Specification
Version	1.0 — Discovery Pass 1 (pre-codebase-audit)
Date	16 July 2026
Prepared by	MindBit Solutions LLP — discovery analysis
System	Guest House Operations Hub · ROOT Platform
Locale	INR · IST · India (GST / DPDP Act jurisdiction)
Companion files	02-SRS.md (source of record) · 02-SRS.html (styled, live diagrams) · full discovery package docs 01–17
Status	DRAFT — pending stakeholder answers to the Question Log (doc 09)

Evidence convention used throughout: **FACT** stated in an authoritative source · **INFERRED** needs confirmation · **REC** analyst recommendation · **OPEN-Q** stakeholder decision required (doc 09).

Contents

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1. Introduction

1.1 Purpose

Specify, to implementation-ready precision, the requirements for the Guest House Operations Hub so that a new engineering team can build, harden, and operate it without further clarification beyond the identified open questions.

1.2 Scope

Covers the operator web application (PWA), its API surface, the AI-agent seam, ingestion/sync integrations (email, iCal), the community network, and the operational envelope (security, performance, backup, monitoring). Excludes the AI assistant's internal runtime (separate service) and live cab dispatch.

1.3 Definitions

Term	Meaning
OTA	Online Travel Agency (Booking.com, Agoda, MakeMyTrip)
Channel	Origin of a booking (Direct, WhatsApp, Website, each OTA); carries a commission %
Travel agent	B2B person bringing bookings for commission; distinct from channel and referral
Stay	Half-open date range [check-in, check-out) stored as Postgres <code>DATERANGE</code>
Block	A room held out of service for a period (maintenance, owner use, iCal-imported busy dates)
Derived availability	Free units = units – overlapping confirmed reservations – overlapping blocks; never a stored counter
HITL	Human-in-the-loop: AI files escalations; a human commits sensitive actions
Seam	The token-gated API surface (<code>/api/agent/*</code> , <code>/api/ingest/*</code>) external services use
Client vs. property	A <i>client</i> (owner business) gets an isolated deployment+DB; one client may run multiple <i>properties</i> inside it
C-Form / Form C	Mandatory registration of foreign guests with Indian authorities (FRRO)
UTR	Unique Transaction Reference for UPI/bank transfers
DPDP	India's Digital Personal Data Protection Act, 2023

1.4 System context

Diagram (rendered in the HTML edition 02-SRS.html; Mermaid source below).

flowchart LR

```

subgraph Client_Deployment["One client = one deployment + one DB (isolated)"]
  HUB["Ops Hub<br/>Next.js 15 + Prisma + Postgres"]
  AGENT["Assistant sidecar<br/>Python · ADK + Gemini · Cloud Run"]
  AGENT -- "token-gated seam<br/>/api/agent/*" --> HUB
end
OWNER["Owner / Reception / Housekeeping<br/>(phone PWA / desktop)"] --> HUB
GUEST["Guest<br/>WhatsApp / chat / voice"] --> AGENT
GMAIL["Owner's inbox<br/>(OTA confirmation emails)"] -- "forwarder →<br/>/api/ingest/email" --> HUB
OTA1["OTA extranets"] -- "iCal import (pull, daily+manual)" --> HUB
HUB -- "iCal export (.ics per room)" --> OTA1
HUB -- "UPI link/QR · invoices · messages" --> GUEST
PEER["Peer properties<br/>(community network)"] -- "grant-gated seam<br/>no PII / occupancy / finance" --> HUB
WA["WhatsApp Cloud API"] -- "adapter, off by default" --> HUB
RZP["Razorpay"] -- "gateway, awaits keys" --> HUB

```

1.5 Constraints (binding)

ID	Constraint	Source
C-01	No OTA extranet scraping/automation; ingest only via owner inbox + official iCal	Hard rule
C-02	No direct OTA connectivity APIs (unavailable to single properties)	Hard rule
C-03	Availability always derived; never a mutable counter	Hard rule
C-04	DB GiST exclusion constraint <code>no_overlapping_confirmed_stays</code> must never be weakened	Hard rule
C-05	No secrets in code; <code>.env</code> git-ignored	Hard rule
C-06	Stack: Next.js 15 App Router, TypeScript strict, Postgres (Supabase), Prisma, Tailwind v4, Zod, Vitest; migrations only via <code>db:migrate:new</code> (Prisma diff engine mangles DATERANGE/GiST)	README/ROADMAP
C-07	One client = one deployment + one database + one agent; nothing shared across clients	ROADMAP
C-08	AI agents never write money or calendar directly; sensitive actions escalate to a human	INTEGRATION design
C-09	Proactive WhatsApp sends require Meta-approved templates (free text only inside 24h window)	Meta policy FACT

2. Product Overview

2.1 Users and roles **FACT**

Role	Access
Owner	Everything, incl. Finance, Pricing, Analytics, Settings, community, multi-property switch
Reception	Bookings, guests, day-to-day ops; no money or setup
Housekeeping	Today + Cleaning only

[GAP] Role model is page/route-level; field-level money masking in API responses is missing (staff-visible responses can carry `grossAmount`) — see NFR-SEC-07, GAP-12.

2.2 Operating environment

Mobile-first PWA (~390 px baseline) installable to home screen; desktop sidebar layout; intermittent-connectivity tolerance claimed (**OPEN-Q** on actual offline depth); reference hosting Vercel + Supabase + GitHub Actions + Vercel Cron; DigitalOcean branch exists.

3. Functional Requirements

Numbering: FR-<MODULE>-<n>. Priority: **Must** / **Should** / **Could** / **Won't** (this release). Status: **?** built (per README/ROADMAP, verify in code) · **✕** missing work / off-by-one detail (inputs/outputs/pre/postconditions/exceptions) is in doc 03 Module Analysis; business rules in doc 05.

3.1 Authentication & user management (AUTH)

ID	Requirement	Pri	Status
FR-AUTH-1	Email+password login with script hashing; session persists ~1 month ("stay logged in")	M	?
FR-AUTH-2	Login rate-limiting 10 attempts/IP/5 min	M	? (per-instance store — see NFR-SEC-05)
FR-AUTH-3	Owner creates staff logins and assigns role + per-property access	M	?
FR-AUTH-4	Self-service password reset (email OTP/link)	M	✕ GAP-10
FR-AUTH-5	User invitation flow (no owner-set passwords)	S	✕ GAP-10
FR-AUTH-6	Session revocation on role change/disable; audit of auth events	M	OPEN-Q
FR-AUTH-7	Optional 2FA for owner accounts	C	✕

3.2 Property & multi-property (PROP)

ID	Requirement	Pri	Status
FR-PROP-1	Property profile: name, address, GST number, check-in/out times, currency, UPI ID, ID-rule strictness (block/warn/off), ID-number-at-booking toggle, ID-scan retention days	M	?
FR-PROP-2	Owner adds properties; switcher appears at 2+; all screens re-scope to acting property	M	?
FR-PROP-3	Guests shared owner-wide; bookings/pricing/finance strictly per-property	M	? (decision 2026-07-15)
FR-PROP-4	Staff property-access editable per user	M	?
FR-PROP-5	All tenant tables carry <u>propertyId</u> ; auto-scoping Prisma extension; raw SQL must scope by hand	M	? (fragile — NFR-SEC-06)

3.3 Rooms & room types (ROOM)

ID	Requirement	Pri	Status
FR-ROOM-1	Room types: name, base rate, max occupancy, rate floor/ceiling	M	?
FR-ROOM-2	Rooms belong to a type; label; archive (retire, keep history); delete only if never booked	M	?
FR-ROOM-3	Amenities per room type (feeds community directory)	S	?

3.4 Availability engine (AVAIL)

ID	Requirement	Pri	Status
FR-AVAIL-1	Availability derived: units – overlapping confirmed reservations – overlapping blocks, per room-type per date	M	?
FR-AVAIL-2	Two confirmed reservations for the same room can never overlap — enforced by Postgres GiST exclusion constraint, not app code	M	?
FR-AVAIL-3	Exclusion-constraint violations surfaced as clean 409 "dates no longer available" (never raw 500)	M	?
FR-AVAIL-4	Cancelled bookings free their dates immediately	M	?
FR-AVAIL-5	Optional oversell safety buffer per room type (don't offer last N units to OTAs/agents)	S	✖ GAP-24 (User Guide advises a manual buffer; no feature exists)

3.5 Booking engine (BOOK)

ID	Requirement	Pri	Status
FR-BOOK-1	Create booking: guest (dedupe by phone), channel, room, dates, arrival time, amount (pre-filled by pricing suggestion), special requests, optional travel agent, ID-confirmation tick	M	?
FR-BOOK-2	Blacklist and scam-number warnings shown at entry; proceed allowed (warn-only)	M	?
FR-BOOK-3	Edit, cancel (frees dates; computes ladder refund),	M	?

ID	Requirement	Pri	Status
	undo of check-in/out steps		
FR-BOOK-4	Statuses: confirmed / cancelled / no_show; check-in/out as timestamps	M	?
FR-BOOK-5	All-bookings list: instant search (guest/phone/room/channel), timeline filters with counts	M	?
FR-BOOK-6	Group/long-stay folio linking multiple room bookings for combined view + billing	S	? (billing depth OPEN-Q)
FR-BOOK-7	Hold-a-room-while-paying (gateway hold with deadline, reusing the exclusion constraint)	S	? awaits Razorpay keys
FR-BOOK-8	Walk-in same-day booking fast path	S	INFERRED
FR-BOOK-9	No-show marking rules (when, by whom, revenue/refund effect)	M	OPEN-Q — status exists; policy undefined

3.6 Calendar (CAL)

Rooms × dates grid; Day/Week/2-Week/Month; colour code vacant/occupied/blocked/conflict/arriving/departing; continuous stay bars with guest+channel; tap-to-book empty nights; date jump; conflict cells red. All ? **FACT** .

3.7 OTA ingestion — email (INGEST)

ID	Requirement	Pri	Status
FR-ING-1	Paste OTA confirmation email → parse guest/dates/amount → staged <u>InboundBooking</u> under Pending review	M	?
FR-ING-2	Review, correct, pick room, create (conflict-checked) or dismiss; original email viewable	M	?
FR-ING-3	Token-gated webhook <u>POST /api/ingest/email</u> + Gmail Apps Script / Cloudflare Worker forwarders for automation; review step retained	M	? off by default
FR-ING-4	Parser tuned per OTA format; unparseable emails staged raw with failure reason	M	? "tuning against real OTA emails remains" FACT
FR-ING-5	Detect OTA modification/cancellation emails and link to the	M	✗ GAP-2 (only confirmations mentioned)

ID	Requirement	Pri	Status
	existing booking rather than staging a duplicate		
FR-ING-6	Duplicate suppression by ota_ref	M	INFERRED

3.8 iCal sync (ICAL)

ID	Requirement	Pri	Status
FR-ICAL-1	Import per-room OTA iCal URL; busy dates become blocks; manual Sync now + daily auto-refresh	M	?
FR-ICAL-2	Export per-room private .ics URL	M	?
FR-ICAL-3	Import failures (URL dead, feed malformed) alert the owner	M	✗ GAP-5
FR-ICAL-4	Removed events in source feed release the corresponding blocks	M	OPEN-Q
FR-ICAL-5	Per-feed tokens (rotate away from single iCal token)	S	✗ (repo's own suggestion)
FR-ICAL-6	Sync frequency configurable / more frequent than daily (lag window = oversell risk)	S	✗ GAP-6

3.9 Guests & CRM (GUEST)

Profiles (name, phone unique, email, address, vehicle, notes, preferences/tags); stays, LTV, repeat badge; reliability score from no-shows + "repeat no-show" badge + shareable alert; ID flags (checked / photocopied / verification complete / consent) gating check-in; blacklist toggle+reason; ID document upload (2 storage env); C-Form section (13 nullable fields: nationality, passport, visa, port+date of entry, purpose); stay history. All ? except noted **FACT**.

ID	Requirement	Pri	Status
FR-GST-1	Guest merge (duplicate records, changed phone)	S	✗ GAP-19
FR-GST-2	DPDP data-subject operations: export & delete a guest's data on request	M	✗ GAP-8
FR-GST-3	C-Form output artefact (print/PDF/CSV in the format the FRRO process accepts)	M	✗ GAP-7 — fields stored, no output OPEN-Q

3.10 Check-in / check-out (CHECK)

Three-stage stay card (not arrived → in-house → checked out) with undo; check-in gated on ID verification (+C-Form for foreigners) per property strictness (block/warn/off); checkout feeds housekeeping. ? **FACT** . **OPEN-Q** : is check-out allowed with balance due? Current behaviour unspecified.

3.11 Housekeeping (HK)

To-clean (post-checkout, "clean first" flag when same-day arrival) and Ready lists; manual "needs cleaning"; assignment to staff with checklist. ? Missing **REC** : inspected/verified state; periodic deep-clean scheduling; block room while dirty option — GAP-20.

3.12 Pricing (PRICE)

Advisory rule engine — weekend, seasons/holidays list, early-bird/last-minute, occupancy/busy-period — compounded then clamped to type floor/ceiling; rate calendar with pinned overrides (dot) and rule-adjusted italics; suggestion pre-fills bookings; never pushed to OTAs; never rewrites saved bookings. ? **FACT** . Rule precedence/compounding order documentation ✖ — **OPEN-Q** .

3.13 Payments (PAY)

ID	Requirement	Pri	Status
FR-PAY-1	Multiple payments per booking; modes cash/UPI/card/bank/OTA-collect; balance due derived	M	?
FR-PAY-2	Advance tracking: advanceRequired + isAdvance payments; status derived	M	?
FR-PAY-3	UPI/bank verification checklist (sender name, funds landed, UTR reference) gating Add payment	M	?
FR-PAY-4	UPI pay link + self-contained QR for outstanding balance (no gateway)	M	?
FR-PAY-5	Razorpay auto-capture + idempotent webhook + room hold	S	? awaits keys
FR-PAY-6	Refund workflow on cancellation per ladder; owner may approve a different amount; refund recorded with status	M	? (payout execution manual INFERRED)
FR-PAY-7	Payment edit/void with audit trail	M	OPEN-Q
FR-PAY-8	Money as integer paise or decimal (replace whole-rupee number math)	M	✖ GAP-9

3.14 Invoicing (INV)

Printable invoice (property name/address/GST no., stay, charges, payments, balance) via browser Print→PDF. ?. Missing for GST compliance **REC** : sequential invoice numbering series, GST rate application per line / tariff slab, SAC code, place of supply, server-side PDF for consistency — GAP-11 **OPEN-Q** .

3.15 Finance (FIN)

Date-ranged tiles (Net to you / Gross / Commission / Outstanding); by-channel breakdown; expenses; balances due; Bookings/Payments CSV. [?](#) **OPEN-Q** : commission basis (gross vs net-of-GST; MMT net-rate vs commission models). **OPEN-Q** : OTA payout reconciliation (recording when the OTA actually pays) **✖** GAP-13.

3.16 Analytics (ANLY)

Occupancy, ADR, RevPAR, average stay, cancellation rate, channel mix; occupancy trend, source-mix donut, revenue-by-channel charts; full CSV export. [?](#).

3.17 Staff / attendance / shifts (STAFF)

Directory (name/role/phone), edit/disable/delete; shift roster; daily attendance present/absent/leave; housekeeping assignment. [?](#). Payroll linkage **✖** (out of scope? **OPEN-Q**).

3.18 Facilities (FAC)

Maintenance requests (priority/assignee/cost, open→in-progress→done); asset register with service-every-N-days + due flag; inventory items (unit, low-stock level, in/out movements, low banner); vendors (+rating); purchase orders draft→ordered→received; vendor payments + summary; tour partners (commission %), tours, tour bookings + revenue/commission summary; drivers + trips (records only). All [?](#).

3.19 Complaints & reviews (CRR)

Complaints: category, priority, assignee, status to resolved with follow-up; filters. Reviews: request tracking + response drafting. [?](#).

3.20 Messaging (MSG)

ID	Requirement	Pri	Status
FR-MSG-1	Outbox logs every outbound message (channel, recipient, text, status)	M	?
FR-MSG-2	Auto-drafts: booking confirmation, pre-arrival (day before), payment reminders	M	? (logged only)
FR-MSG-3	WhatsApp Cloud API adapter flips status to sent/failed when enabled	M	? needs credentials + approved templates
FR-MSG-4	Template management + params through the seam; language variants (Khasi/Hindi/English)	M	✖ GAP-3
FR-MSG-5	Inbound guest replies visible to the operator (two-way thread)	S	✖ GAP-4 — outbox only OPEN-Q

3.21 AI agent seam & escalations (AGENT)

ID	Requirement	Pri	Status
FR-AGT-1	Token-gated (AGENT_TOKEN , fail-closed) endpoints: availability, quote, create	M	?

ID	Requirement	Pri	Status
	reservation (same GiST transaction, 409 on overlap), queue message, file escalation		
FR-AGT-2	Escalation queue: source/category/severity tags, guest original message + AI summary, open→in-progress→resolved; linked booking actions done via normal screens	M	?
FR-AGT-3	Owner-editable runtime policies (Settings → Assistant rules), applied ~1 min, no redeploy	M	?
FR-AGT-4	Guest/owner persona isolation; canonical security block outranks owner policies; model fallback chain; per-turn diagnostics	M	? (in sidecar)
FR-AGT-5	Escalation SLA / notification to owner's phone when an escalation arrives	M	✗ GAP-14 (badge only; no push)
FR-AGT-6	Property-aware agent requests; booking derives property from room	M	?

3.22 Community network (COMM)

Connect via code; per-peer share switches (rooms, referrals, scam numbers, bad-guest alerts, vendors, drivers); directory by amenities; overflow referrals with accept→book (conflict-checked)→attributed revenue→derived reciprocal-credit ledger; verified scam/bad-guest sharing (evidence required, dispute/appeal, auto-expiry, hashed number matching); shared vendor/driver lists; all opt-in; never shares guest PII, occupancy, or finance; single audited grant-gated seam. ? **FACT** .

ID	Requirement	Pri	Status
FR-COM-1	Credit-ledger settlement/dispute mechanism (what happens when balances grow one-sided)	S	OPEN-Q
FR-COM-2	Network governance: who moderates disputed reports across independent deployments; where does the shared registry live	M	OPEN-Q — cross-deployment architecture unclear INFERRED

3.23 Settings, import, audit (SET)

All settings sections per User Guide table ?; CSV import of past guests/bookings (conflict-checked per row) ?; audit log of sensitive actions (cancellations, refunds, blacklisting, user & consent changes) ?. **REC** Audit coverage should extend to: payment edits, price overrides, settings changes, community shares, ID views/downloads — GAP-15.

3.24 Offline & sync (OFFLINE)

User Guide claims: offline changes saved and synced on reconnect; conflicting offline bookings surfaced. Depth unverified **OPEN-Q** — the honest spec (queue? cache-only? which screens?) must come from code audit; treat as **?**.

3.25 Localization (L10N)

ID	Requirement	Pri	Status
FR-L10N-1	Operator console i18n framework; Khasi + English UI (Hindi S)	S→M for inclusion goals	✗ GAP-16 INFERRED
FR-L10N-2	Guest-facing artefacts (messages, invoices) language selection	S	✗
FR-L10N-3	AI assistant Khasi/Hindi/English voice+text	M	? (sidecar) FACT

4. Non-Functional Requirements

4.1 Security

ID	Requirement	Status
NFR-SEC-01	Passwords script-hashed; sessions HTTP-only; secrets via env only	?
NFR-SEC-02	All external seams token-gated, fail-closed	?
NFR-SEC-03	Cross-client isolation absolute (separate deployments/DBs)	? by architecture
NFR-SEC-04	In-client tenant isolation: Prisma auto-scoping + hand-scoped raw SQL	? but fragile — add Postgres RLS as defence-in-depth (GAP-12)
NFR-SEC-05	Rate limiting on shared store (Upstash/Redis) — per-instance memory is weak on serverless	✗ known debt
NFR-SEC-06	Lint/CI guard for raw SQL missing <code>property_id</code> scope	✗ REC
NFR-SEC-07	Field-level authorization: money fields masked server-side for non-owner roles	✗ known debt
NFR-SEC-08	ID documents: private bucket, signed URLs, access logged, encrypted at rest	? partial (adapter exists; logging unverified)
NFR-SEC-09	Community hashed-number matching resistant to trivial re-identification (peppered/keyed hash — phone numbers are low-entropy; plain hashing is reversible by enumeration)	OPEN-Q REC
NFR-SEC-10	Dependency and secret scanning in CI; security headers (CSP etc.)	OPEN-Q unverified

4.2 Privacy & compliance

ID	Requirement	Status
NFR-PRV-01	DPDP 2023: lawful-purpose notice, recorded consent (exists ?), data-principal rights (access/correction/erasure ✗), breach notification procedure ✗ , retention enforcement (purge cron exists but inert until `idRetentionDays` set — default should be a sane value, not indefinite REC)	Partial — GAP-8
NFR-PRV-02	Form C: capture ? ; artefact/submission support ✗	GAP-7
NFR-PRV-03	GST-compliant invoices (numbering, tax lines)	GAP-11
NFR-PRV-04	Cross-border processing disclosure (Gemini/US cloud) in guest-facing notice	✗ REC
NFR-PRV-05	Data-processing terms between MindBit and each client (who is fiduciary vs processor)	OPEN-Q

4.3 Performance & capacity

ID	Requirement	Target REC
NFR-PRF-01	Interactive screens on mid-range Android over 3G/4G	P75 < 2.5 s LCP
NFR-PRF-02	Booking create round-trip	< 1.5 s P95
NFR-PRF-03	Calendar month view, 20 rooms	< 2 s P95
NFR-PRF-04	Scale envelope per client	≤ 5 properties, ≤ 60 rooms, ≤ 30k reservations/yr — small by design; N+1s acceptable only inside this envelope
NFR-PRF-05	Agent seam availability/quote	< 800 ms P95 (conversation latency budget)

4.4 Availability & resilience

ID	Requirement	Status
NFR-AVL-01	Target 99.5% monthly availability per client REC	Undefined today
NFR-AVL-02	Documented, tested backup & restore: daily automated backup, RPO ≤ 24h, RTO ≤ 4h, quarterly restore drill	✗ GAP-1 (Supabase backups exist platform-side; procedure undocumented/untested INFERRED)
NFR-AVL-03	Graceful degradation when sidecar/AI is down (hub fully usable; escalations queue)	? by architecture INFERRED
NFR-AVL-04	Cron failure (iCal sync, purge) detection	✗ GAP-5

ID	Requirement	Status
	& alert	

4.5 Observability & support

ID	Requirement	Status
NFR-OBS-01	Error tracking (e.g. Sentry), structured logs, uptime checks per client deployment	✗ GAP-17
NFR-OBS-02	Fleet dashboard: version, health, cron status across all client deployments	✗ GAP-18 — critical for 25+ clients
NFR-OBS-03	Support model: channel, hours, SLA, in-app "report a problem"	OPEN-Q

4.6 Usability & accessibility

ID	Requirement	Status
NFR-UX-01	Mobile-first ~390 px; thumb-reachable primary actions; dark/light; density options	?
NFR-UX-02	Usable by non-technical operators; plain-language errors	? intent; validate in pilot UAT
NFR-UX-03	WCAG 2.1 AA for core flows (contrast, touch targets ≥44 px, screen-reader labels)	✗ unassessed — GAP-21
NFR-UX-04	Low-literacy affordances (icons, voice via assistant)	REC future

4.7 Maintainability & delivery

ID	Requirement	Status
NFR-MNT-01	CI: lint + typecheck + Vitest + agent pytest on PR	?
NFR-MNT-02	Route-level test for create-overlap→409; housekeeping derivation test	✗ repo's own top gap
NFR-MNT-03	Safe-migration discipline documented & enforced (CI check that migrations came from the helper REC)	Partial
NFR-MNT-04	Fleet upgrade path: migrations run on deploy (prisma migrate deploy) ?; coordinated multi-client rollout tooling ✗ GAP-18	

5. External Interfaces

Interface	Direction	Protocol	Notes
Owner inbox → hub	In	Email forwarder → HTTPS webhook (token)	Gmail Apps Script or Cloudflare Email Worker

Interface	Direction	Protocol	Notes
			FACT
OTA iCal	In/Out	HTTPS .ics pull/serve	Binary busy/free, per room, hours-level lag; not offered on all listing types FACT
WhatsApp Cloud API	Out (in future?)	HTTPS	Template approval required; 24h session window FACT
Razorpay	In (webhook) / Out	HTTPS, idempotent webhook	Awaits merchant keys FACT
UPI deep link / QR	Out	upi:// URI + SVG QR	No gateway; verification is manual checklist FACT
Agent sidecar	In	HTTPS, AGENT_TOKEN	availability/quote/reserve/message/escalate FACT
Peer deployments (community)	In/Out	Grant-gated seam	Exact cross-deployment topology OPEN-Q
Browser print	Out	CSS print	Invoices; server-side PDF deferred

6. Data model

See doc 12 (entity catalogue + Mermaid ERD reflecting the 55-model schema). Correctness core: `Reservation.stay DATERANGE` + GiST exclusion (`WHERE status='confirmed'`), `Block.period DATERANGE`; availability computed, never stored.

7. Error handling principles

- Constraint violations → domain-specific 4xx with friendly copy (409 for overlap) — never raw 500 **FACT**.
- Consistent `{ data }` / `{ error }` envelope **FACT**.
- Parser failures stage the raw email rather than dropping it (FR-ING-4).
- Seam auth failures fail closed with no information leak.
- **REC** Add: global error boundary with user-visible incident ID; retry semantics documented per endpoint; webhook idempotency keys (Razorpay planned; extend to ingest).

8. Open items blocking final sign-off

The **OPEN-Q** tags above; consolidated with rationale in doc 09. The five that block the most design surface: Q-FIN-01 (commission basis), Q-LEG-02 (Form C artefact), Q-TEC-04 (offline truth), Q-LEG-03 (community topology), Q-BUS-03 (monetisation model).